

```
%python
import oml

df_train = oml.sync(query="SELECT * FROM B2B_USER.VW_TRAIN_CAPA1")
features = df_train.drop('RUC')

km_model = oml.km(n_clusters=4)
print("Entrenando modelo...")
km_model.fit(features)
print("Entrenamiento completado!")

try:
    oml.drop(model_name='B2B_KMEANS_CAPA1')
except:
    pass

km_model.save(name='B2B_KMEANS_CAPA1', description='Modelo K-Means Capa 1')
print("Modelo guardado como B2B_KMEANS_CAPA1.")
```

Entrenando modelo...

Entrenamiento completado!

Fail to execute line 16: km_model.save(name='B2B_KMEANS_CAPA1', description='Modelo K-Means Capa 1')

Traceback (most recent call last):

```
File "/tmp/python707121301058488970/zeppelin_python.py", line 270, in <module>
    exec(code, _zcUserQueryNameSpace)
    ~~~^~~~~~
```

```
File "<stdin>", line 16, in <module>
```

AttributeError: 'km' object has no attribute 'save'

```
%sql
SELECT MODEL_NAME, ALGORITHM, MINING_FUNCTION
FROM USER_MINING_MODELS
ORDER BY CREATION_DATE DESC
```



MODEL_NAME	ALGORITHM	MINING_FUNCTION
B2B_KMEANS_CAPA1	KMEANS	CLUSTERING

Página 1 de 1 (1 de 1 elementos)

1

 Cargar más

```
%sql
SELECT RUC,
        CLUSTER_ID(B2B_KMEANS_CAPA1 USING *) AS CLUSTER_ASIGNADO
FROM B2B_USER.VW_FEATURES_ML_CAPA1
WHERE ROWNUM <= 10
```



>

Introducir búsqueda

RUC 	CLUSTER_ASIGNADO 
190301850001	3
190159892001	3
190167062001	3
190167100001	3
190004163001	3

Página

1

 de 2 (1-5 de 10 elementos)



1

2



Cargar más

```
%sql
SELECT CLUSTER_ID(B2B_KMEANS_CAPA1 USING *) AS CLUSTER_ID,
       COUNT(*) AS TOTAL_EMPRESAS
FROM B2B_USER.VW_FEATURES_ML_CAPA1
WHERE ROWNUM <= 50000
GROUP BY CLUSTER_ID(B2B_KMEANS_CAPA1 USING *)
ORDER BY CLUSTER_ID
```



>

Introducir búsqueda

CLUSTER_ID ↕	TOTAL_EMPRESAS ↕
3	30167
5	47
6	19304
7	482

```
%sql
SELECT f.RUC,
       CLUSTER_ID(B2B_KMEANS_CAPA1 USING *) AS CLUSTER_OML,
       c.CLUSTER_ID AS CLUSTER_LOCAL,
       c.SEGMENTO AS SEGMENTO_LOCAL
FROM B2B_USER.VW_FEATURES_ML_CAPA1 f
JOIN B2B_USER.ML_CLUSTERS_CAPA1 c ON f.RUC = c.RUC
ORDER BY CLUSTER_OML, CLUSTER_LOCAL
```



Introducir búsqueda

RUC ↕	CLUSTER_OML ↕	CLUSTER_LOCAL ↕	SEGMENTO_LOCAL ↕
990010153001	3	0	Comercio formal mad
1790018156001	3	0	Comercio formal mad
190316890001	3	1	Industriales consolida
1768185570001	3	2	Comercio formal eme
1793194826001	3	2	Comercio formal eme

Cargar más

```
%script
BEGIN
  DBMS_DATA_MINING.DROP_MODEL('B2B_KMEANS_CAPA1');
END;
```



PL/SQL procedure successfully completed.

```
%python
import oml

df_train = oml.sync(query="SELECT * FROM B2B_USER.VW_TRAIN_CAPA1_OHE")
features = df_train.drop('RUC')

km_model = oml.km(n_clusters=4)
print("Entrenando con OHE manual...")
km_model.fit(features)
print("Completado!")
```



Entrenando con OHE manual...
Completado!

```
%script
DECLARE
  v_name VARCHAR2(128);
BEGIN
  SELECT MODEL_NAME INTO v_name FROM USER_MINING_MODELS WHERE ROWNUM = 1;
  DBMS_DATA_MINING.RENAME_MODEL(v_name, 'B2B_KMEANS_CAPA1');
END;
```



PL/SQL procedure successfully completed.

```
%sql
SELECT CLUSTER_ID(B2B_KMEANS_CAPA1 USING *) AS CLUSTER_OML,
       c.CLUSTER_ID AS CLUSTER_LOCAL,
       c.SEGMENTO AS SEGMENTO_LOCAL,
       COUNT(*) AS TOTAL
FROM B2B_USER.VW_FEATURES_ML_CAPA1_OHE f
JOIN B2B_USER.ML_CLUSTERS_CAPA1 c ON f.RUC = c.RUC
GROUP BY CLUSTER_ID(B2B_KMEANS_CAPA1 USING *), c.CLUSTER_ID, c.SEGMENTO
ORDER BY CLUSTER_OML, CLUSTER_LOCAL
```



Introducir búsqueda

CLUSTER_OML	CLUSTER_LOCAL	SEGMENTO_LOCAL
3	0	Comercio formal maduro en Pichincha
3	1	Industriales consolidados de alta afinidad
3	2	Comercio formal emergente en Pichincha
4	0	Comercio formal maduro en Pichincha
4	2	Comercio formal emergente en Pichincha

Cargar más

```
%script
BEGIN
  DBMS_DATA_MINING.DROP_MODEL('B2B_KMEANS_CAPA1');
END;
```



PL/SQL procedure successfully completed.

```
%python
import oml

df_train = oml.sync(query="SELECT * FROM B2B_USER.VW_TRAIN_CAPA1_FINAL")
features = df_train.drop('RUC')

# Apagamos el Auto Data Preparation (PREP_AUTO) para que actúe como Scikit-Learn
km_model = oml.km(n_clusters=4, PREP_AUTO='OFF')

print("Entrenando matriz idéntica a Python...")
km_model.fit(features)
print("Completado!")
```



Entrenando matriz idéntica a Python...
Completado!

```
%script
DECLARE
  v_name VARCHAR2(128);
BEGIN
  SELECT MODEL_NAME INTO v_name FROM USER_MINING_MODELS ORDER BY CREATION_DATE DESC FETCH
  DBMS_DATA_MINING.RENAME_MODEL(v_name, 'B2B_KMEANS_CAPA1');
END;
```



PL/SQL procedure successfully completed.

```
%sql
SELECT CLUSTER_ID(B2B_KMEANS_CAPA1 USING *) AS CLUSTER_OML,
       c.CLUSTER_ID AS CLUSTER_LOCAL,
       c.SEGMENTO AS SEGMENTO_LOCAL,
       COUNT(*) AS TOTAL
FROM B2B_USER.VW_FEATURES_ML_CAPA1_FINAL f
JOIN B2B_USER.ML_CLUSTERS_CAPA1 c ON f.RUC = c.RUC
GROUP BY CLUSTER_ID(B2B_KMEANS_CAPA1 USING *), c.CLUSTER_ID, c.SEGMENTO
ORDER BY CLUSTER_OML, CLUSTER_LOCAL
```



Introducir búsqueda

CLUSTER_OML 	CLUSTER_LOCAL 	SEGMENTO_LOCAL 
3	0	Comercio formal maduro en Pichincha
3	1	Industriales consolidados de alta afinidad
5	0	Comercio formal maduro en Pichincha
5	1	Industriales consolidados de alta afinidad
5	2	Comercio formal emergente en Pichincha

Página

1

 de 3 (1-5 de 12 elementos)



1

2

3



Cargar más

```
%python
import oml
df_train = oml.sync(query="SELECT * FROM B2B_USER.VW_TRAIN_CAPA2_FINAL")
features = df_train.drop('RUC')

km_model = oml.km(n_clusters=4, PREP_AUTO='OFF')
print("Entrenando modelo Capa 2...")
km_model.fit(features)
print("Completado Capa 2!")
```



Entrenando modelo Capa 2...
Completado Capa 2!


```
%script
DECLARE
  v_name VARCHAR2(128);
BEGIN
  SELECT MODEL_NAME INTO v_name FROM USER_MINING_MODELS ORDER BY CREATION_DATE DESC FETCH
  DBMS_DATA_MINING.RENAME_MODEL(v_name, 'B2B_KMEANS_CAPA2');
END;
```



PL/SQL procedure successfully completed.

```
%script
GRANT SELECT ON MINING MODEL B2B_KMEANS_CAPA2 TO B2B_USER;
```



Grant succeeded.
